

Chapter 7

1. What does it mean for a wireless network to be operating in “infrastructure mode”? If the network is not in infrastructure mode, what mode of operation is it in, and what is the difference between that mode of operation and infrastructure mode?
2. In 4G/5G networks, what effect will handoff have on end-to-end delays of datagrams between the source and destination?
3. Suppose the IEEE 802.11 RTS and CTS frames were as long as the standard DATA and ACK frames. Would there be any advantage to using the CTS and RTS frames? Why or why not?
4. What is the role of the base station in 4G/5G cellular architecture? With which other 4G/5G network elements (mobile device, MME, HSS, Serving Gateway Router, PDN Gateway Router) does it directly communicate with in the control plane? In the data plane?
5. Does the LTE wireless access network use FDMA, TDMA, or both? Explain your answer.
6. a) What is meant by a “visited network” and a “home network” in 4G/5G cellular architecture?
b) What is meant by “hand over” of a network device?
7. When a mobile device is handed over from one base station to another in a 4G/5G network, which network element makes the decision to initiate that handover? Which network element chooses the target base station to which the mobile device will be handed over?
8. Consider the single-sender CDMA example in Figure 7.5. What would be the sender’s output (for the 2 data bits shown) if the sender’s CDMA code were (1, 1, -1, 1, 1, -1, -1, 1)?

Chapter 8

1. Suppose you want to encrypt the message 10010111 by encrypting the decimal number that corresponds to the message. What is the decimal number?
2. In what way does a hash provide a better message integrity check than a checksum (such as the Internet checksum)?
3. What does it mean for a signed document to be verifiable and nonforgeable?
4. Suppose Alice has a message that she is ready to send to anyone who asks. Thousands of people want to obtain Alice's message, but each wants to be sure of the integrity of the message. In this context, do you think a MACbased or a digital-signature-based integrity scheme is more suitable? Why?
5. What is the de facto e-mail encryption scheme? What does it use for authentication and message integrity?
6. Suppose Bob initiates a TCP connection to Trudy who is pretending to be Alice. During the handshake, Trudy sends Bob Alice's certificate. In what step of the TLS handshake algorithm will Bob discover that he is not communicating with Alice?
7. Consider RSA with $p = 7$ and $q = 13$.
 - a. What are n and z ?
 - b. Let e be 17. Why is this an acceptable choice for e ?
 - c. Find d such that $de = 1 \pmod{z}$.
 - d. Encrypt the message $m = 9$ using the key (n, e) . Let c denote the corresponding ciphertext. Show all work.
8.
 - a. Using RSA, choose $p = 5$ and $q = 7$, and encode the numbers 12, 19, and 27 separately. Apply the decryption algorithm to the encrypted version to recover the original plaintext message.
 - b. Choose p and q of your own and encrypt 1834 as one message m .